



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI ANJANEYA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

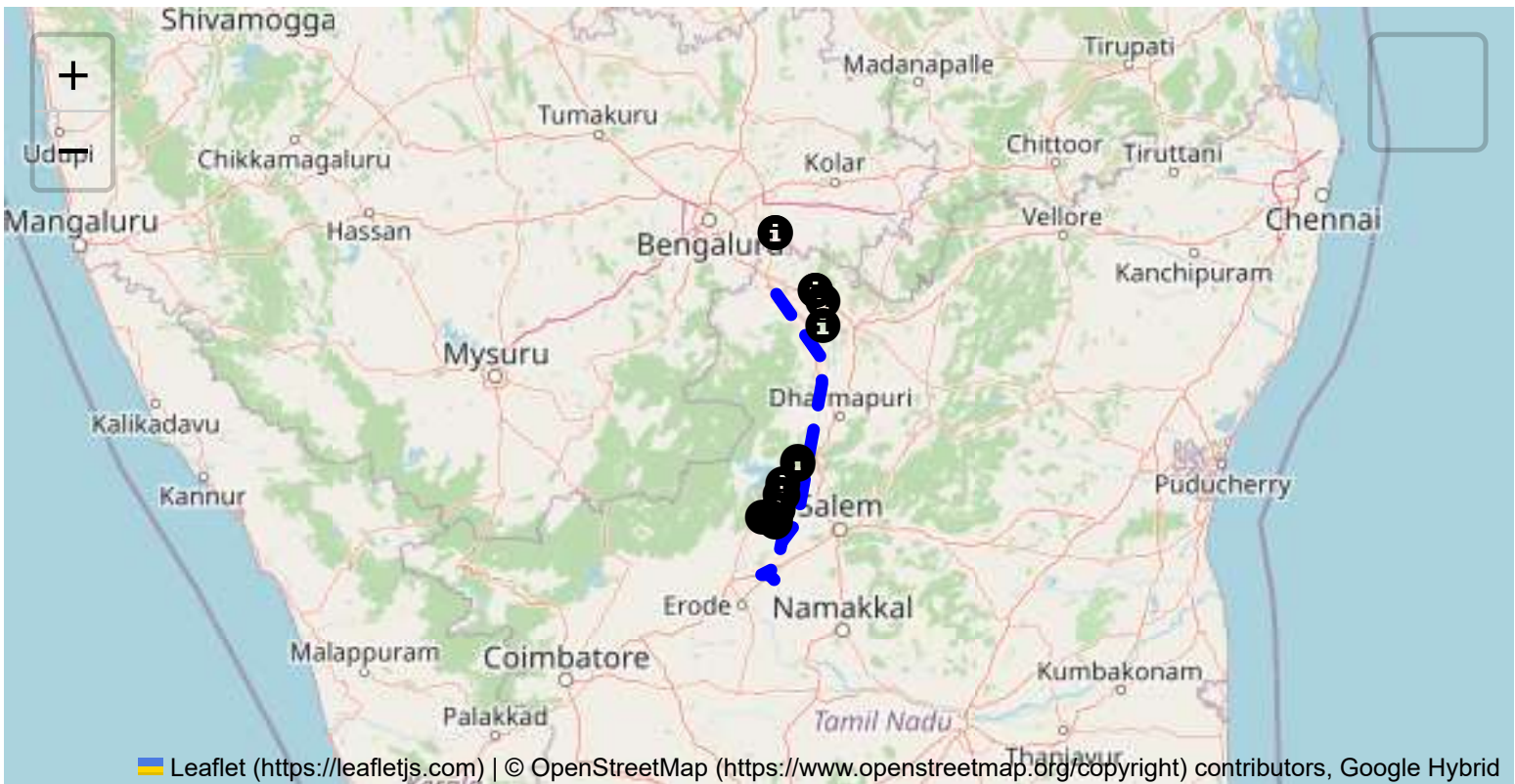
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

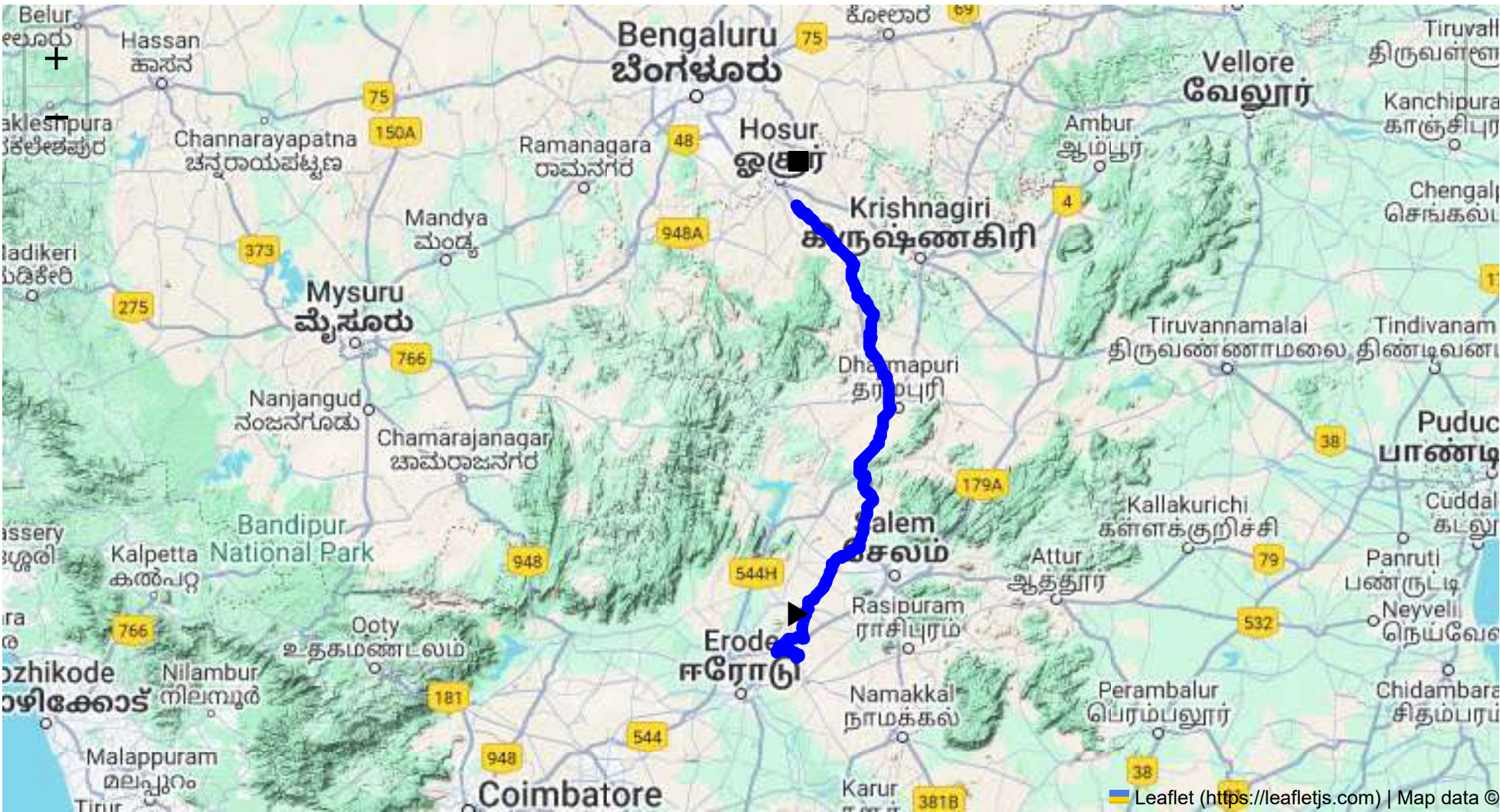
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 180.06 km
Estimated Duration: 3.9 hours
Adjusted Duration (Heavy Vehicle): 4.9 hours
Start: (11.4381, 77.8734)
End: (12.671942, 77.870503)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Universal Kitchen Equipment in Hosur covers approximately 180.06 kilometers and traverses various highways and state roads, including SH 86 and SH 17. The journey moves through a combination of rural and urban landscapes.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a tropical climate with hot summers (March to June) and substantial rainfall during the monsoon (July to September). The winter season (November to February) is relatively mild. Heavy rains during the monsoon can lead to flooding in low-lying areas, causing roadblocks or landslides, especially on hilly terrains near Salem and Dharmapuri.

3. Analysis of Traffic Patterns

Traffic is notably dense in urban areas like Salem and the vicinity of Hosur, with peak hours typically between 8-10 AM and 5-8 PM. Rural stretches may have minimal congestion but watch for slow-moving agricultural vehicles. Expect increased traffic on weekends and holidays, especially near Salem and other townships en route.

4. Assessment of Road Quality and Infrastructure

- **Sangagiri to Salem:** Generally well-maintained roads with good signage but may encounter occasional potholes.
- **Salem to Palacode:** Road quality varies; some stretches may have uneven surfaces due to ongoing infrastructure works.
- **Final leg to Hosur:** Narrower roads in some patches might require cautious navigation, particularly during rainy conditions.

5. Suggestions for Alternative Routes for Emergencies

For emergency detours, consider the NH44, which runs parallel in some sections. This national highway offers faster travel and better facilities but may require additional kilometers and bypass these points of congestion.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu has stringent regulations for the transport of hazardous materials. Ensure all vehicles carry appropriate documentation, labels, and adhere to weight limitations. The use of GPS tracking for real-time monitoring is recommended.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Salem:** Historical data indicates occasional traffic accidents involving heavy vehicles, often due to congestion or unfavorable weather.
- **Hosur:** Known for industrial activity, there have been past incidents of hazardous material spillage, primarily due to inadequate containment.

8. Environmental Considerations and Sensitive Areas

- **Forested Areas:** Sections near Palacode and certain stretches approaching Hosur may pass through environmentally sensitive zones, including protected forests. Exercise caution to prevent spills or environmental damage.
- **Water Bodies:** Proximity to rivers may increase flood risks during monsoon.

9. Analysis of Communication Coverage

Most sections of the route have adequate mobile network coverage, but there might be dead zones in rural or hilly areas, particularly between Tharamangalam and Palacode.

10. Estimated Emergency Response Times for Different Route Segments

- **Sangagiri to Salem:** Emergency response may take approximately 30-45 minutes.
- **Salem to Palacode:** Expect a response time of 45 minutes to one hour due to variable terrain.
- **Palacode to Hosur:** Response times may improve to 30-45 minutes near urban centers.

12. Overall Summary of Risk Assessment

The route is moderately safe, with major risks arising from weather conditions, variable road quality, and congestion near urban centers. Plan travel by factoring in potential delays during rainfalls and city traffic. Maintain compliance with hazardous material transport regulations and have an emergency plan for detours or breakdowns.

Preparation, vigilance, and regular communication with control centers are crucial, especially in areas with patchy mobile coverage.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	Medium	11.43995, 77.87431	30 KM/Hr
2	Turn	High	11.44107, 77.87528	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	Medium	11.45352, 77.85698	30 KM/Hr
5	Turn	Medium	11.45764, 77.85219	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	High	11.45571, 77.81702	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	Medium	11.45654, 77.81706	30 KM/Hr
9	Turn	Medium	11.46916, 77.82431	30 KM/Hr
10	Blind Spot	Blind Spot	11.49225, 77.89560	10 KM/Hr
11	Turn	High	11.49453, 77.88488	15 KM/Hr
12	Turn	Medium	11.56348, 77.90077	30 KM/Hr
13	Turn	Medium	11.57547, 77.90567	30 KM/Hr
14	Turn	Medium	11.59189, 77.90645	30 KM/Hr
15	Turn	Medium	11.59323, 77.90796	30 KM/Hr
16	Turn	Medium	11.63857, 77.94831	30 KM/Hr
17	Turn	High	11.64207, 77.95199	15 KM/Hr
18	Turn	Medium	11.64708, 77.95119	30 KM/Hr
19	Turn	High	11.69596, 77.97122	15 KM/Hr
20	Turn	Medium	11.71721, 77.99982	30 KM/Hr
21	Turn	Medium	11.73772, 78.03606	30 KM/Hr
22	Turn	High	11.73665, 78.03907	15 KM/Hr
23	Turn	Medium	11.74313, 78.04724	30 KM/Hr
24	Turn	Medium	11.77961, 78.05775	30 KM/Hr
25	Turn	Medium	11.86658, 78.08267	30 KM/Hr
26	Turn	Medium	11.87053, 78.08372	30 KM/Hr
27	Turn	Medium	11.87966, 78.08187	30 KM/Hr
28	Turn	Medium	11.91211, 78.05852	30 KM/Hr
29	Turn	Medium	11.92609, 78.06513	30 KM/Hr
30	Turn	Medium	11.93334, 78.06409	30 KM/Hr
31	Turn	Medium	11.93719, 78.05372	30 KM/Hr
32	Turn	Medium	11.95457, 78.04685	30 KM/Hr
33	Turn	Medium	11.95688, 78.04803	30 KM/Hr
34	Turn	High	11.95828, 78.04983	15 KM/Hr
35	Turn	Medium	11.95997, 78.04946	30 KM/Hr
36	Turn	High	11.96408, 78.05131	15 KM/Hr
37	Turn	High	11.96511, 78.04966	15 KM/Hr
38	Turn	High	11.96697, 78.04959	15 KM/Hr
39	Turn	Medium	11.96819, 78.05088	30 KM/Hr
40	Turn	Medium	12.12010, 78.13282	30 KM/Hr
41	Turn	Medium	12.20466, 78.12205	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
42	Turn	Medium	12.26535, 78.08429	30 KM/Hr
43	Turn	High	12.29603, 78.07411	15 KM/Hr
44	Turn	High	12.29651, 78.07545	15 KM/Hr
45	Turn	Medium	12.31084, 78.07863	30 KM/Hr
46	Turn	Medium	12.31250, 78.07771	30 KM/Hr
47	Turn	High	12.31355, 78.07486	15 KM/Hr
48	Turn	High	12.37264, 78.08517	15 KM/Hr
49	Turn	High	12.37599, 78.08366	15 KM/Hr
50	Turn	High	12.37835, 78.08521	15 KM/Hr
51	Turn	Medium	12.37961, 78.08436	30 KM/Hr
52	Turn	Medium	12.38660, 78.07509	30 KM/Hr
53	Turn	Medium	12.40540, 78.06757	30 KM/Hr
54	Turn	Medium	12.41089, 78.06163	30 KM/Hr
55	Turn	Medium	12.41275, 78.06102	30 KM/Hr
56	Turn	Medium	12.41714, 78.05554	30 KM/Hr
57	Turn	Medium	12.41944, 78.04947	30 KM/Hr
58	Blind Spot	Blind Spot	12.42512, 78.04656	10 KM/Hr
59	Turn	Medium	12.42461, 78.04641	30 KM/Hr
60	Turn	High	12.42277, 78.04470	15 KM/Hr
61	Turn	High	12.42368, 78.04310	15 KM/Hr
62	Turn	Medium	12.42736, 78.04412	30 KM/Hr
63	Blind Spot	Blind Spot	12.51381, 78.03208	10 KM/Hr
64	Turn	Medium	12.51395, 78.02897	30 KM/Hr
65	Turn	High	12.51632, 78.02571	15 KM/Hr
66	Turn	Medium	12.52916, 78.02035	30 KM/Hr
67	Turn	Medium	12.60997, 77.93650	30 KM/Hr
68	Turn	Medium	12.65368, 77.88750	30 KM/Hr
69	Turn	Medium	12.65636, 77.88560	30 KM/Hr
70	Turn	Medium	12.65803, 77.88260	30 KM/Hr
71	Turn	Medium	12.67159, 77.87294	30 KM/Hr
72	Blind Spot	Blind Spot	12.67271, 77.87086	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
3	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
4	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
5	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
6	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
7	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
8	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
9	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
10	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
11	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
12	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
13	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
14	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
15	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
16	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
22	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
23	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
24	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
26	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
27	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
28	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	hospital	Government Hospital Palacode	12.2964824, 78.0761735	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
19	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
29	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
31	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium
32	school	Government Higher Secondary School	12.5844162, 77.9669764	30 km/h	Medium
33	school	Panchayat Union Primary School	12.6078768, 77.9381865	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44107, 77.87528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45654, 77.81706



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49225, 77.89560



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49453, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56348, 77.90077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57547, 77.90567



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59189, 77.90645



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59323, 77.90796



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63857, 77.94831



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64207, 77.95199



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64708, 77.95119



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69596, 77.97122



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71721, 77.99982



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73772, 78.03606



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73665, 78.03907



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74313, 78.04724



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77961, 78.05775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86658, 78.08267



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87053, 78.08372



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87966, 78.08187



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.91211, 78.05852



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92609, 78.06513



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93334, 78.06409



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93719, 78.05372



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95457, 78.04685



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95688, 78.04803



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95828, 78.04983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95997, 78.04946



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96408, 78.05131



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96511, 78.04966



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96697, 78.04959



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96819, 78.05088



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12010, 78.13282



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.20466, 78.12205



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26535, 78.08429



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.29603, 78.07411



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.29651, 78.07545



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.31084, 78.07863



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.31250, 78.07771



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.31355, 78.07486



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.37264, 78.08517



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.37599, 78.08366



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.37835, 78.08521



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.37961, 78.08436



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.38660, 78.07509



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40540, 78.06757



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.41089, 78.06163



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.41275, 78.06102



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.41714, 78.05554



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.41944, 78.04947



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.42512, 78.04656



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.42461, 78.04641



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.42277, 78.04470



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.51381, 78.03208



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.51395, 78.02897



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.51632, 78.02571



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52916, 78.02035



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.60997, 77.93650



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.65803, 77.88260



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.67159, 77.87294



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.67271, 77.87086
